

6 Neurotransmitters are involved in the transmission of a nerve impulse across a synapse.

(a) Which chemical would increase the likelihood of a nerve impulse being initiated at a post-synaptic membrane?

(1)

- ☐ **A** auxin
- ☐ **B** dopamine
- ☐ **C** gibberellin
- ☐ **D** statin

(b) Heart muscle cells obtain over 95% of their energy from oxidative phosphorylation.

When heart muscle cells are cultured in a low oxygen concentration they die.

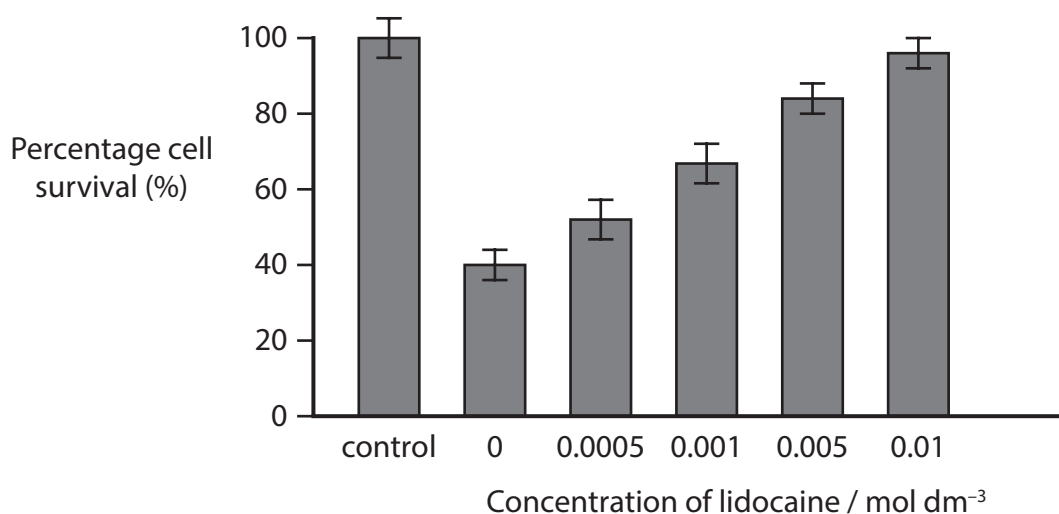
It has been suggested that lidocaine can be used to protect heart muscle cells from this cell death.

Lidocaine works by blocking the movement of sodium ions through the membranes surrounding the neurones. This prevents the initiation and conduction of impulses along the neurone.

In an investigation the effect of lidocaine on the cell survival of heart muscle cells cultured in a low oxygen concentration was investigated.

The control culture was given an excess of oxygen and no lidocaine. The other cultures had a low oxygen concentration and different concentrations of lidocaine.

The results are shown in the graph.



(i) Assess the effects of lidocaine on cell survival.

(3)

(ii) Heart muscle cells are myogenic.

Which describes the term **myogenic**?

(1)

- ☐ **A** a hormone that initiates differentiation
- ☐ **B** cardiac muscle that is able to contract with no external stimulus
- ☐ **C** how a heart beat is initiated in the atrioventricular node
- ☐ **D** how muscles work in pairs

(iii) Suggest how lidocaine protects heart muscle cells from effects of low oxygen.

(2)

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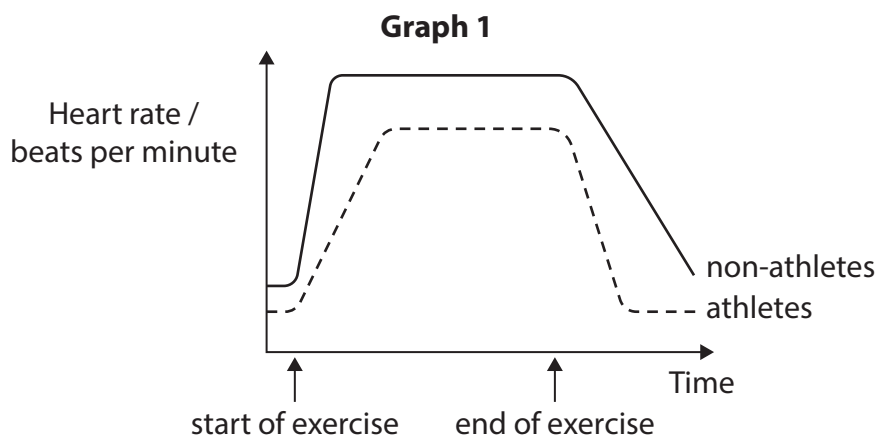
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*(c) Cardiovascular disease (CVD) is one of the biggest causes of death in the world. In 2021, nearly 25% of all deaths were attributed to CVD.

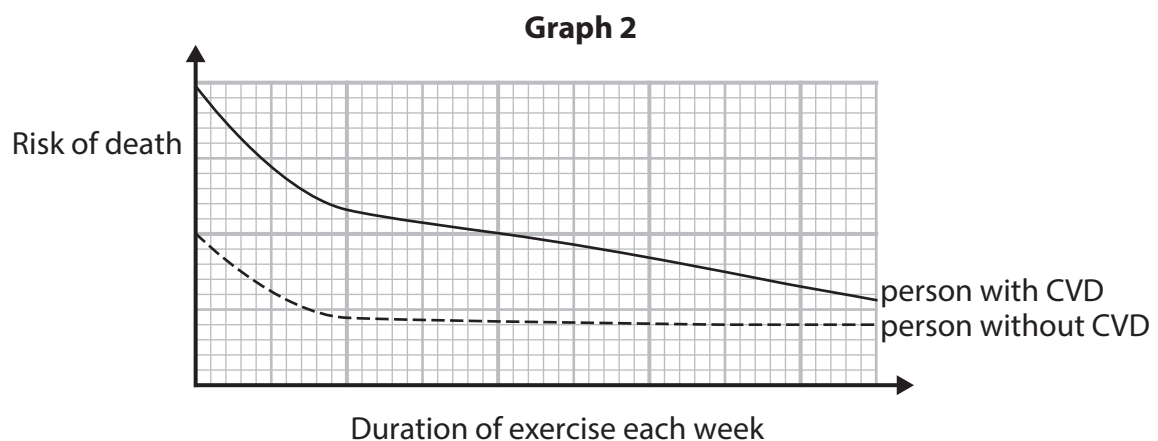
Exercise can help prevent the onset and development of CVD.

In an investigation, the heart rates of athletes and non-athletes were measured before, during and after exercise.

Graph 1 shows the relationship between heart rate and exercise for non-athletes and athletes.



Graph 2 shows how exercise can reduce the risk of death depending on whether the person has CVD or not.



Discuss how exercise reduces the risk of death from CVD.

Use the information in the graphs and your own knowledge to support your answer.

(6)

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(Total for Question 6 = 13 marks)